
Project Report - Form-A

AGRINOVA

Turning crop residue into income - not smoke, not drain waste

Predict. Protect. Prosper. Recycle.

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Affiliation	NCSC / NCERT Children's Science Congress stream
Language of report	English
Field area	Madhyamgram and Barasat
District / State	North 24 Parganas, West Bengal
Comparison belt	Punjab (Patiala, Kharar, Nabha)
Sample size	n = 100 farm households (stratified, representative)
Live desk	https://agrinova-hazel.vercel.app
Guide teacher	

Abstract

AgriNova is a Class XI project under the NCSC theme Science and Innovation for Sustainability. It asks why rice straw still leaves the farm as smoke in Punjab and as mixed drain waste in Madhyamgram and Barasat, when compost pads, paper mills and biomass gates already buy the same material. The missing step is a shared listing - quantity, moisture, neighbourhood - that a smallholder, a plant gate and a ward desk can all see.

The study has two parts. Part I defines rice straw from CRM, SWM Rules 2016, and air-quality notes, and places the work on India's long-horizon net-zero pathway (~2070). Part II is a representative field survey of $n = 100$ farm households: 35 in Madhyamgram, 35 in Barasat, and 30 in Punjab (phone), stratified by landholding band. AgriNova (<https://agrinova-hazel.vercel.app>) was then demonstrated on those clusters.

Results. Burned 24%; stacked/dumped 40%; mixed waste 20%; sold 16%. 81 of 100 would list if a trolley came in five days (~650.4 t ~ Rs 455,280 at Rs 700/t). Carbon uses 1.5 tCO₂e per tonne as a working estimate, not a certified credit. Farmer-first carbon lines accrue to the household that diverted the lot. The gap is not another scheme pamphlet - it is moisture-aware matching, a visible pickup date, and a ward total of utilised versus dumped tonnes without farmer phones on the officer screen.

Keywords

Rice straw; Madhyamgram; Barasat; Punjab; waste management; transparency; farmer-first carbon; India 2070; NCSC Form-A; representative sample.

Why this project?

Burned or dumped straw is wasted biomass, dirty air, blocked drains, and zero farm cash. From Kendriya Vidyalaya Dum Dum the walkable field is Madhyamgram and Barasat in North 24 Parganas. Punjab enters by phone (Patiala, Kharar, Nabha) as the published burn clock. Four angles guided every sheet and every screen.

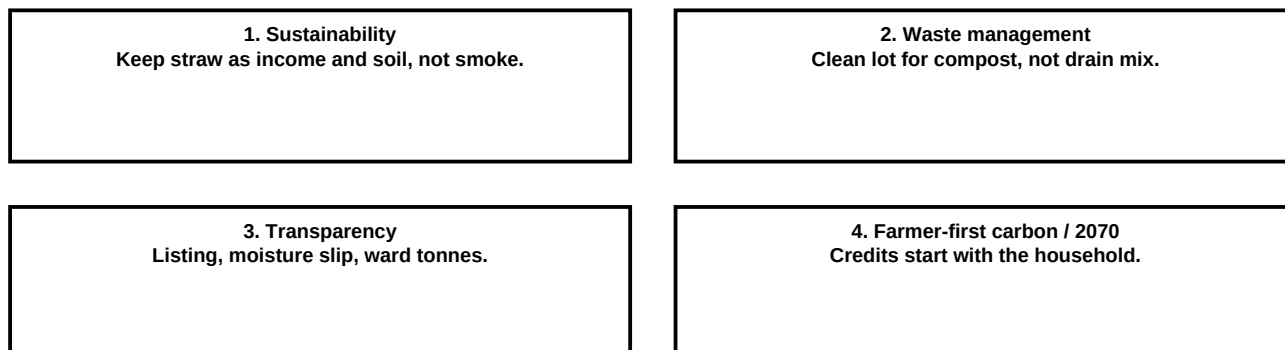


Figure 1. Four research angles used throughout the study.

1. Sustainability of farming - keep organic matter and income on the farm side of the ledger.
2. Waste management - SWM Rules, 2016 already ask urban bodies to process biodegradable waste; straw must arrive as a clean lot, not mixed drain matter.
3. Transparency in management - one listing, one moisture slip, one pickup date, one ward total; no hidden broker cut; no farmer phone on the officer screen.
4. Farmer-first carbon toward India 2070 - estimated avoided CO₂e starts with the household that changed behaviour, as a local contribution to the national net-zero horizon, without claiming certified national accounting.

India 2070 as a cluster pathway

This project does not close a national carbon budget. It shows what a ward can actually run: fewer open burns, fewer mixed drain loads, more utilised tonnes, a farmer-visible gate price, and an estimated carbon line. Practical abatement in Madhyamgram and Barasat is how a large goal becomes local habit.

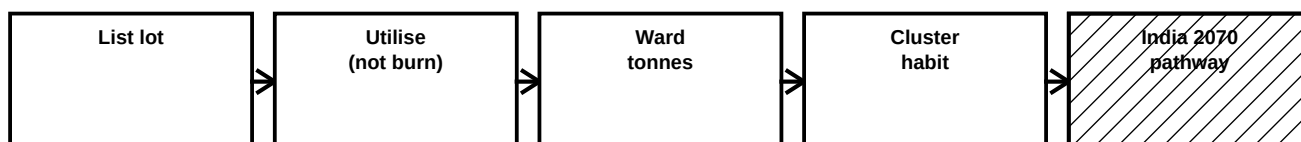


Figure 8. Cluster pathway - local tonnes toward India's long-horizon net-zero (~2070).

1. Selection of the problem

The focal theme asks for science that makes a local waste stream less harmful. The stream we can reach is peri-urban paddy leftover in Madhyamgram and Barasat: wet stacks after rain, drain push, and mixing into household waste that conservancy then lifts as mixed matter. Punjab (Patiala belt) is the comparison stratum because that is the short sowing-window burn clock. One listing must serve both failure modes.

We selected this problem because (i) the material is the same - rice straw; (ii) Madhyamgram and Barasat fail mainly by dump/mix while Punjab fails by burn; (iii) a compost pad already exists while drains still take mixed straw - a transparency failure, not only a technology failure; (iv) India 2070 needs cluster habits; (v) a Class XI team can close a representative sample of 100 households.

1.1 Hypothesis

H1 (primary). If a smallholder can list last season's straw with a moisture note and is promised pickup inside five days on a shared desk, then selling or gifting will replace burning or dumping as the default, provided net gate price after freight is positive.

H2. The main barrier is absence of an own plant phone, not absence of scheme pamphlets. H3. Wet Madhyamgram / Barasat lots match compost-first; dry Punjab lots match biomass moisture caps. H4. Ward officers will accept utilised-vs-dumped totals if farmer phones stay off the screen. H5. Farmer-first estimated carbon increases willingness to list compared with buyer-only claims.

Null idea we tried to reject: that smallholders do not care about residue. The survey tests whether care appears when the management object (listing + trolley date + moisture slip) exists.

1.2 Variables

- Independent idea: moisture-aware listing with five-day pickup (willingness + desk demonstration).
- Dependent idea: residue fate (burn / stack-dump / mix / sell) and willingness to list.
- Stratifiers: place (Madhyamgram / Barasat / Punjab) and landholding band S / M / L.
- Working constants: 2.0 t/acre rice straw; Rs 700/t gate; 1.5 tCO₂e/t avoided (estimate).

1.3 Objectives

1. Record residue fate across n = 100 representative farm households.
2. Name the main barrier by place and landholding band.
3. Check whether plant staff will lift 2-6 acre lots if moisture is declared.
4. Document what ward registers capture versus what transparency requires.
5. Field-test AgriNova and propose a ward-level abatement system.

1.4 Need and relevance

Madhyamgram and Barasat are peri-urban North 24 Parganas, next to our school. Conservancy already lifts mixed biodegradable waste; small leftover piles never become a specified lot. Food, Agriculture and Health: burning and drain dumping damage air, hygiene and soil. Sustainability: circular straw is income plus abatement. India 2070: wards that can show utilised tonnes can grow climate habit.

2. Basic information about the element chosen

The element is rice straw (paddy residue), sometimes with wheat stubble. It is an agricultural by-product, a municipal biodegradable-waste risk, and a climate-relevant biomass stream.

- Quantity: published residue-to-grain band about 1.8-2.4 t/acre; working figure = 2.0 t/acre.
- Use: biomass energy, paper furnish, compost / biomethanation - each with a moisture spec.
- Burning: near-zero farm cash; particulate load on the winter airshed (literature context).
- Dumping: wrong specification for compost pads; becomes a conservancy complaint in Madhyamgram or Barasat.
- Carbon: about 1.5 tCO₂e avoided per tonne utilised instead of open burned (educational estimate).

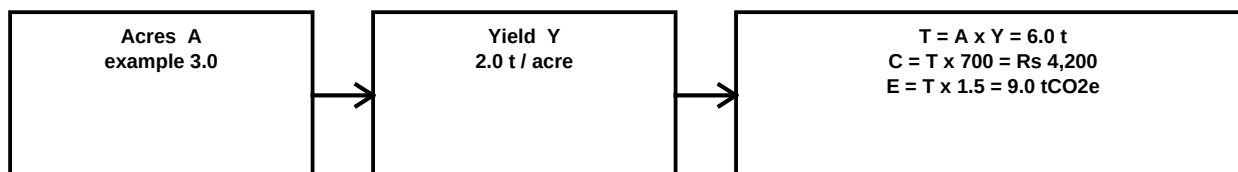


Figure 4. Working mathematics used on the desk and in this book (Doltala example).

Element	Why it matters	How recorded
Residue fate	Burn / dump / sell / mix	One choice
Harvest-sow window	Clock that makes burning rational	Four bands
Own buyer phone	Matching / transparency gap	Yes / broker / none
Main barrier	Cause, not slogan	One choice
5-day pickup	Product test	Yes / maybe / no
Landholding band	Economic strata	S / M / L acres
Register contents	What the ward already sees	Officer note
Language	Inclusion	Bangla / Punjabi / Hindi / English

Table 1. Data elements fixed before fieldwork.

3. Previous studies and policy context

We did not invent stubble burning or eastern residue waste. A short cited shelf defines the problem before the 100 sheets.

Policy and waste management

NPMCR (2014) asks states to map surplus and move it in-situ or to industry. CRM machinery is densest in the north-west burn belt. SWM Rules, 2016 require segregation and processing of biodegradable waste, including agricultural leftover that enters the urban stream - directly relevant to Madhyamgram and Barasat. NGT orders raise the cost of fire; they do not create a buyer phone in the village.

Air, volume and farm economics

SAFAR / IITM and CPCB episode notes are the public source for peak-day PM_{2.5} from residue fire. CRM notes round Punjab paddy straw to the order of tens of millions of tonnes a kharif. MSP cards explain why grain is sold and straw is treated as a nuisance. Biomass tenders often sit near Rs 600-1200/t. A 3-acre pile at 2.0 t/acre is about 6 t; at Rs 700/t that is about Rs 4,200 before freight.

Climate horizon

India's net-zero ambition around 2070 frames why cluster abatement matters. AgriNova positions farmer-first estimated carbon as a behavioural lever inside that horizon, while refusing certified national-credit claims. NCSC / NCERT guidance also frames why sample size and representative design matter: a handful of convenient sheets can mislead.

What the shelf does not give

A transparent place where a 3-acre farmer in Doltala (Madhyamgram), a compost pad, and a Ward 21 Barasat councillor share one lot without putting KCC numbers on a portal. That absence is the problem we measured.

4. Analysis of the problem and measurement

The problem has three clocks and one transparency gap that cuts across all three.

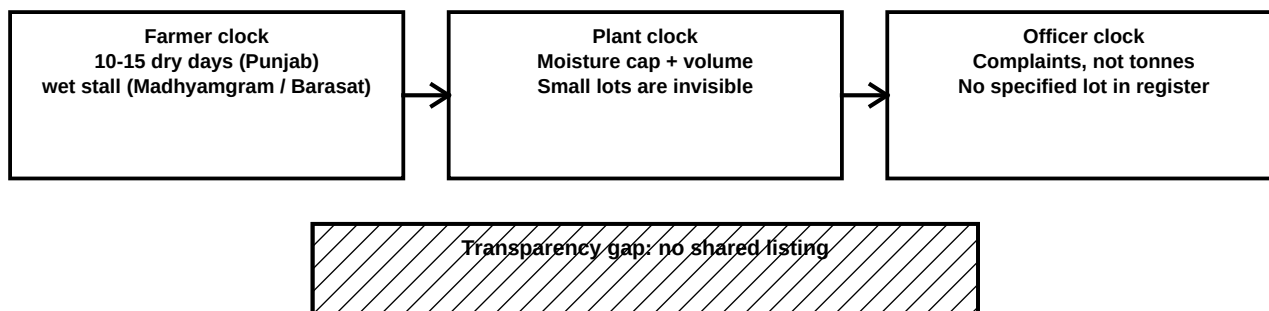


Figure 2. Three clocks and one missing object - the shared listing.

1. Farmer clock - 10-15 dry days in Punjab; longer but wetter days in Madhyamgram and Barasat.
2. Plant clock - large tonnage at a moisture cap; a two-acre pile is invisible unless aggregated.
3. Officer clock - mixed-waste complaints, not tonnes that left as a specified lot.
4. Transparency gap - broker rumour and wet-test fights; even sold rows can be unfair.

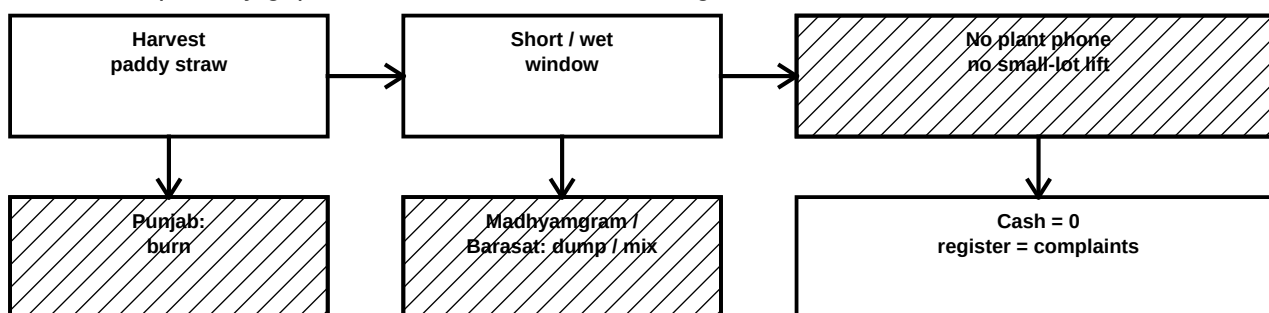


Figure 6. Causal sketch used to design the questionnaire. Hatched = failure modes.

4.1 Suggested solutions (before field testing)

1. Machines only - without listed lots, balers still arrive late.
2. Fines only - raise the cost of burning; do not create a trolley or a slip.
3. Scheme PDF / helpline - local sheets asked for Bangla voice and a pickup slot.
4. Matching desk with farmer-first carbon (chosen) - one listing, one score, one trolley, one ward total.

5. Survey design - area, duration, method

Survey is the key to accurate data collection, interpretation and conclusions. NCSC guidance is clear: the type and number of questions, and the size of the sample, need care. A very small sample risks wrong conclusions. This study closed a minimum of 100 farm households using representative (stratified) sampling.

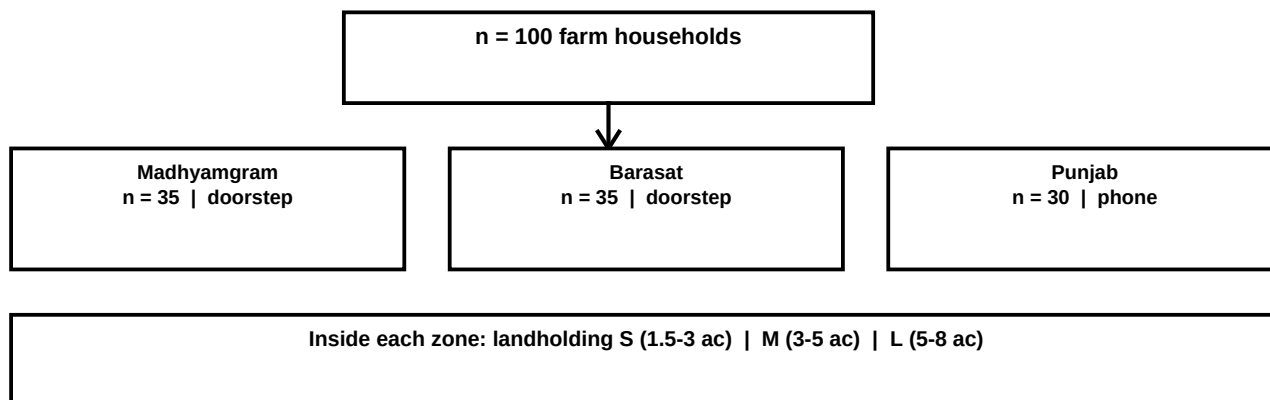


Figure 3. Representative sample - place x landholding (economic proxy).

Stratum	n	Share	Rationale
Madhyamgram (doorstep)	35	35%	Home dump/mix pocket
Barasat (doorstep)	35	35%	Ward / drain interface
Punjab phone (Patiala belt)	30	30%	Published burn clock
Small 1.5-3 ac (S)	40	40%	Mills often skip
Medium 3-5 ac (M)	34	34%	Typical peri-urban lot
Larger 5-8 ac (L)	26	26%	Can fill a trolley

Table 2. Sampling frame (n = 100).

Duration: September 2025. Doorstep work started at Hridaypur (Barasat) and continued through Doltala, Noapara, Michael Nagar, Nabapally and Ward 21. Punjab calls (Kharar, Nabha, Ghanaur, Rajpura) were made the same month. Rules: 1.5-8 acre land; options read aloud; one sentence in the speaker's words; no Aadhaar / KCC / mobile stored.

Supporting conversations (design, not sample inflation): Madhyamgram compost pad; Punjab biomass cabin; Councillor, Ward 21, Barasat; Sanitary Inspector, Madhyamgram. Statistical claims rest on the 100 farm sheets.

6. Questionnaire / data collected

6.1 Farmer instrument (n = 100)

1. Fate of most of last season's straw - sold / stacked-dumped / burned / mixed waste.
2. Harvest to next sowing - under 10 / 10-15 / 16-25 / more than 25 days.
3. Own plant phone - yes / only broker / none.
4. Main barrier if not sold - no phone / too wet / transport / unaware / I did sell.
5. Would you list if a trolley came in five days? - yes / maybe / no.
6. Preferred language - Bangla voice / Punjabi / Hindi / English.
7. Operated acres and one sentence in your words.

6.2 Buyer sheet (supporting)

- Plant type and moisture cap; mixed drain straw refused?
- Do 2-6 acre lots reach this gate without a broker?
- Would a listing with moisture help you send one trolley?

6.3 Official sheet (supporting)

- What goes into the register today?
- Biggest gap between leftover and the pad?
- Would an aggregate utilised view (no farmer phones) help the weekly note?

Instruments were kept short after a long schedule was refused at the first Madhyamgram sitting. Options were read aloud. Mode (doorstep or phone) was marked on every farmer sheet.

7. Activities taken up

1. Mapped sustainability, waste management, transparency and India 2070 onto a local problem.
2. Closed n = 100 farmer sheets in Madhyamgram, Barasat and Punjab.
3. Ran supporting plant-floor and ward conversations.
4. Built AgriNova: West Bengal ward defaults; State / District / Ward login; explainable match; farmer-first carbon; Kisan Bandhu leaf samples with known results.
5. Demonstrated dump-type and burn-type listings; interpreted data by stratum.

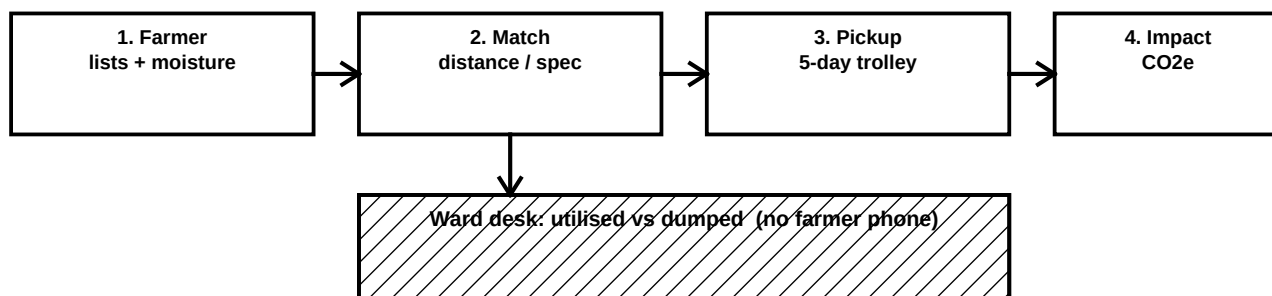


Figure 7. AgriNova abatement loop tested after the survey.

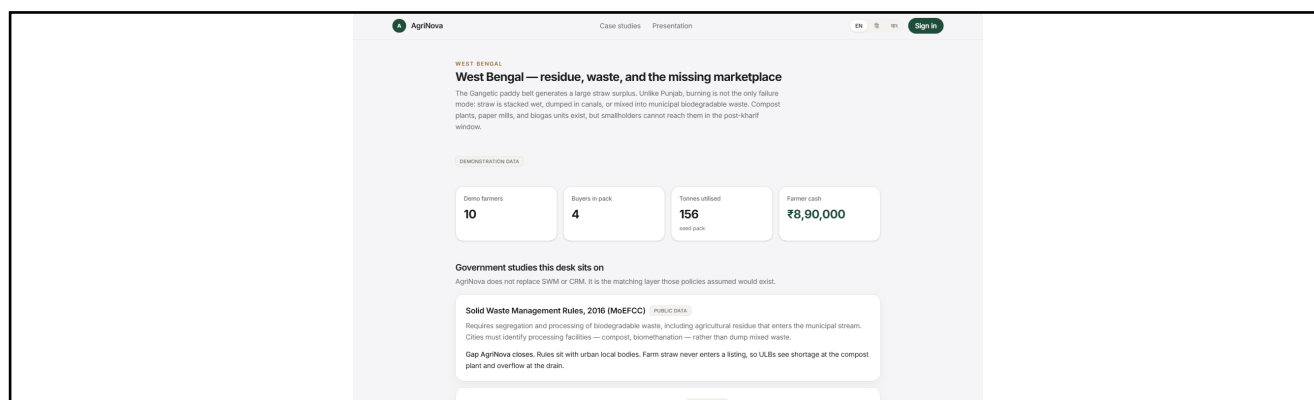


Figure 11. AgriNova landing desk used in the demonstration.

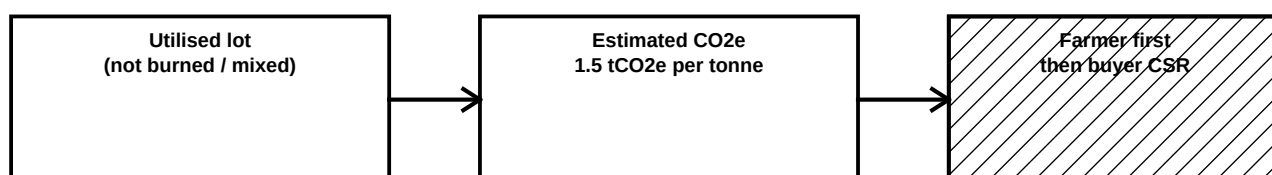


Figure 5. Farmer-first carbon line (educational estimate, not a certified credit).

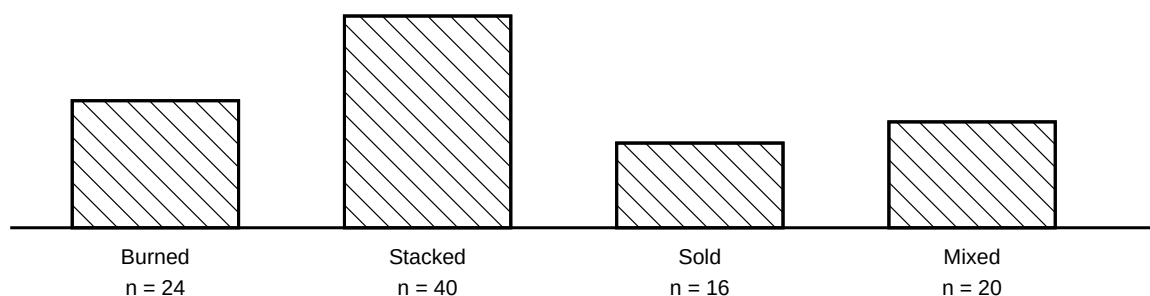
8. Analysis of data

8.1 Residue fate (n = 100)

Fate	Madhyamgram	Barasat	Punjab	All	Share
Burned	0	0	24	24	24%
Stacked / dumped	20	20	0	40	40%
Sold / gifted	5	5	6	16	16%
Mixed waste	10	10	0	20	20%

Table 3. Last-season residue fate by place.

Figure 9. Residue fate (n = 100)



Hatched bars remain readable on a black-and-white printout.

8.2 Matching gap and product test

Own plant phone 7; broker 18; none 75. Five-day pickup: Yes 81, Maybe 19. Bangla voice dominates Madhyamgram and Barasat sheets; Punjabi dominates the Punjab calls. Bengal fails after monsoon wetness; Punjab fails on a dry twelve-day clock. The shared hole is the listing.

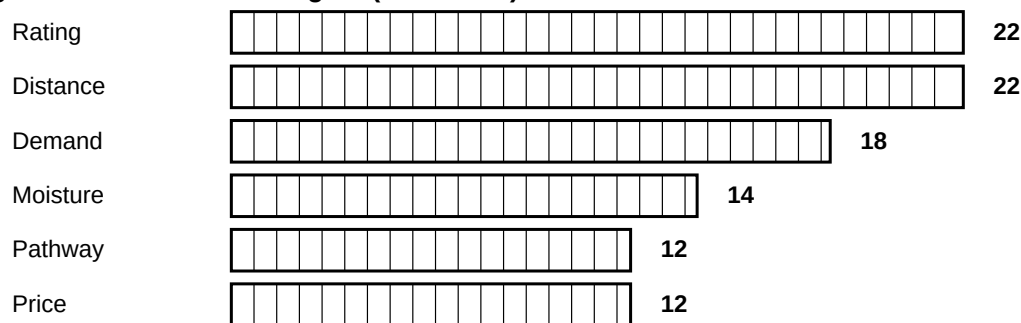
8.3 Fate by landholding band

Band	n	Burned	Stacked	Sold	Mixed	Yes 5-day
S 1.5-3 ac	40	10	16	6	8	32
M 3-5 ac	34	8	14	6	6	28
L 5-8 ac	26	6	10	4	6	21

Table 4. Small lots need neighbourhood pooling.

8.4 Sample impact (working constants)

- Land total 395.0 ac -> estimated straw 790.0 t at 2.0 t/acre.
- Yes-to-list acres 325.2 -> ~650.4 t -> ~Rs 455,280 at Rs 700/t.
- Burned 94.3 ac -> ~188.6 t -> ~282.9 tCO₂e. Stacked/mixed 236.4 ac -> ~472.8 t still outside clean SWM.
- If Yes lots utilise: ~975.6 tCO₂e avoided (estimate).

Figure 10. Match-score weights (sum = 100)

Moisture can zero a bad match - transparency before the truck leaves the lane.

8.5 Hypothesis check

H1 is supported at willingness: 81/100 say yes to five-day pickup. Last-season fate still shows burn (24) and dump/mix (60) - why the listing is needed. H2 is supported by the none-phone share. H3 is supported by plant specs and AgriNova match. H4 is supported by councillor / SI notes. H5 is a design principle on the desk, not a randomised trial.

8.6 Voices that shaped the desk

Hridaypur (Barasat): wet heap, too small alone - maybe if neighbours pool. Doltala (Madhyamgram): no mill number; will stack clean if given a date. Ward 21 (Barasat): leftover to drain; wants Bangla, not a PDF. Noapara: sold via broker, moisture cut - wants the slip before the truck leaves. Kharar (Punjab): twelve days for wheat; will not burn if a trolley date is visible. Madhyamgram pad: hungry for clean feedstock; refuses mixed drain straw. Ward 21 / Madhyamgram SI: utilised-vs-dumped totals, no farmer phones.

8. Demonstration on AgriNova

Default login and case study are West Bengal first (Madhyamgram and Barasat). A wet Madhyamgram-type lot matches short-haul compost; a dry Punjab-type lot matches a $\leq 15\%$ biomass cabin. Kisan Bandhu ships three curated leaf samples (rice blast, bacterial blight, wheat rust) with known educational results.

The screenshot shows the AgriNova login interface. It includes a 'Login' button and a 'Select your role to continue' dropdown. The roles listed are Farmer (with a sub-option 'Set farmer details'), Buyer (with a sub-option 'Purchase biomass'), Government (with a sub-option 'AI & policy back'), and Admin (with a sub-option 'Operations: 1800'). Below this, there is a form for 'Ramesh Singh - Smallholder - 8 acres' with fields for Name, District (Patna), Village / office (Kharar), Phone (+91 98765 10421), Land (acres) (5), Age (42), and Gender (Male). A 'Login as Ramesh' button is at the bottom.

Figure 12. Login: Identity / State-Ward / Land. Default: Madhyamgram and Barasat.

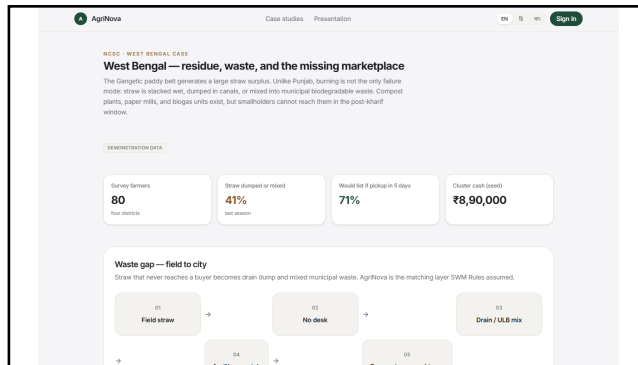


Figure 13. West Bengal case-study hub.

The screenshot shows the 'List residue — Ramesh Singh' interface. It includes a 'Farm & land' section with fields for Farmer (Ramesh Singh), Phone (+91 98765 10421), Location (Kharar, Patna, Punjab), Total land (acres) (5), Area under this residue (acres) (2.5), and Source crop (Rice). Below this, there is a 'Detected residue' section with a 'Continue' button. At the bottom, there are buttons for 'Rice Straw', 'Wheat Stubble', 'Cotton Stalks', 'Sugarcane Tops', and 'Maize Stover'.

Figure 14. Sell / match desk - transparent score.

The screenshot shows the 'Kisan AI' interface. It includes a 'Soil profile' section with fields for Soil type (Alluvial), pH (7.2), Organic carbon % (0.45), and Nitrogen status (Medium). Below this, there is a 'KisanSathi' section with a 'OpenAI' button. At the bottom, there is a 'KisanSathi' section with a 'Type or tap the mic' button.

Figure 15. Kisan Bandhu - listing advice and leaf scan.

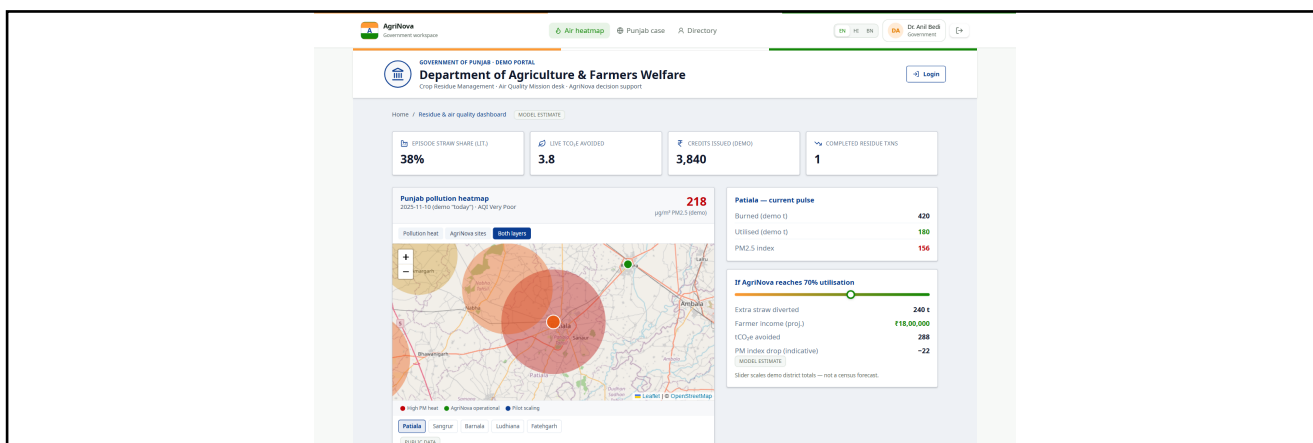


Figure 16. Ward desk - utilised vs dumped, no farmer phones.

8. Survey roster (n = 100)

Complete closed sample. Mode: D = doorstep, P = phone. Fate: B burn, K stack/dump, S sold, M mixed. Band: S/M/L. 5d: Y yes, M maybe. Places are Madhyamgram, Barasat, or Punjab (Kharar, Nabha, Ghanaur, Rajpura, Samana, Patiala).

#	Name	Place	St	How	Band	Ac	Fate	Phone	5d
1	Ramesh Das	Doltala	WB	D	S	1.6	K	none	M
2	Sukumar Roy	Noapara	WB	D	S	1.7	K	none	Y
3	Anil Ghosh	Michael Nagar	WB	D	S	1.8	M	none	Y
4	Tapas Mondal	Madhyamgram	WB	D	S	1.9	K	none	Y
5	Kamal Halder	Udayrajpur	WB	D	S	2.0	M	none	Y
6	Arun Bhattachary	Doltala	WB	D	S	2.1	S	broker	Y
7	Biplab Banerjee	Noapara	WB	D	S	2.2	K	none	M
8	Sanjay Mitra	Michael Nagar	WB	D	S	2.3	M	none	Y
9	Debashis Sarkar	Madhyamgram	WB	D	S	2.4	K	none	Y
10	Pradip Sen	Udayrajpur	WB	D	S	2.5	S	broker	Y
11	Gopal Biswas	Doltala	WB	D	S	2.6	K	none	Y
12	Nirmal Dutta	Noapara	WB	D	S	2.7	K	none	Y
13	Ashok Mondal	Michael Nagar	WB	D	S	2.8	M	none	M
14	Bikash Ghosh	Madhyamgram	WB	D	S	2.9	K	none	Y
15	Subrata Pal	Udayrajpur	WB	D	M	3.2	K	none	M
16	Manoranjan Das	Doltala	WB	D	M	3.3	K	none	Y
17	Chandan Roy	Noapara	WB	D	M	3.5	M	none	Y
18	Partha Chatterje	Michael Nagar	WB	D	M	3.6	K	none	Y
19	Sourav Bhowmik	Madhyamgram	WB	D	M	3.8	M	none	Y
20	Tapan Seal	Udayrajpur	WB	D	M	3.9	S	broker	Y
21	Ajit Saha	Doltala	WB	D	M	4.1	K	none	M
22	Milan Kundu	Noapara	WB	D	M	4.2	M	none	Y
23	Bimal Halder	Michael Nagar	WB	D	M	4.4	K	none	Y
24	Dipak Naskar	Madhyamgram	WB	D	M	4.5	S	broker	Y
25	Haradhan Ghosh	Udayrajpur	WB	D	M	4.7	K	none	Y
26	Kartik Das	Doltala	WB	D	M	4.8	K	none	Y
27	Nemai Mondal	Noapara	WB	D	L	5.3	K	none	M
28	Prafulla Roy	Michael Nagar	WB	D	L	5.6	K	none	Y
29	Ranjan Biswas	Madhyamgram	WB	D	L	5.9	M	none	Y
30	Sudhir Adhikari	Udayrajpur	WB	D	L	6.2	K	none	Y
31	Uttam Ghosh	Doltala	WB	D	L	6.5	M	none	Y
32	Alok Banerjee	Noapara	WB	D	L	6.8	S	broker	Y
33	Barun Dey	Michael Nagar	WB	D	L	7.1	K	none	M
34	Chittaranjan Sen	Madhyamgram	WB	D	L	7.4	M	none	Y
35	Dilip Mukherjee	Udayrajpur	WB	D	L	7.7	K	none	Y
36	Goutam Pal	Hridaypur	WB	D	S	1.6	K	none	M
37	Himangshu Das	Nabapally	WB	D	S	1.7	K	none	Y
38	Indrajit Roy	Ward 21	WB	D	S	1.8	M	none	Y
39	Jaharlal Mondal	Barasat	WB	D	S	1.9	K	none	Y
40	Kalyan Ghosh	Jessore Road	WB	D	S	2.0	M	none	Y
41	Loknath Biswas	Hridaypur	WB	D	S	2.1	S	broker	Y
42	Madhab Chandra	Nabapally	WB	D	S	2.2	K	none	M
43	Narayan Seal	Ward 21	WB	D	S	2.3	M	none	Y

#	Name	Place	St	How	Band	Ac	Fate	Phone	5d
44	Pankaj Dutta	Barasat	WB	D	S	2.4	K	none	Y
45	Rabin Sarkar	Jessore Road	WB	D	S	2.5	S	broker	Y
46	Satyajit Bose	Hridaypur	WB	D	S	2.6	K	none	Y
47	Tarun Mitra	Nabapally	WB	D	S	2.7	K	none	Y
48	Utpal Banik	Ward 21	WB	D	S	2.8	M	none	M
49	Vivekananda Das	Barasat	WB	D	S	2.9	K	none	Y
50	Yudhisthir Roy	Jessore Road	WB	D	M	3.2	K	none	M
51	Amitava Ghosh	Hridaypur	WB	D	M	3.3	K	none	Y
52	Basudeb Mondal	Nabapally	WB	D	M	3.5	M	none	Y
53	Chinmoy Das	Ward 21	WB	D	M	3.6	K	none	Y
54	Dulal Roy	Barasat	WB	D	M	3.8	M	none	Y
55	Ekkari Biswas	Jessore Road	WB	D	M	3.9	S	broker	Y
56	Falguni Saha	Hridaypur	WB	D	M	4.1	K	none	M
57	Ganesh Adhikari	Nabapally	WB	D	M	4.2	M	none	Y
58	Haripada Naskar	Ward 21	WB	D	M	4.4	K	none	Y
59	Iswar Chandra	Barasat	WB	D	M	4.5	S	broker	Y
60	Jagadish Pal	Jessore Road	WB	D	M	4.7	K	none	Y
61	Kanchan Dutta	Hridaypur	WB	D	M	4.8	K	none	Y
62	Lalmohan Ghosh	Nabapally	WB	D	L	5.3	K	none	M
63	Mihir Banerjee	Ward 21	WB	D	L	5.6	K	none	Y
64	Nikhil Seal	Barasat	WB	D	L	5.9	M	none	Y
65	Purna Chandra	Jessore Road	WB	D	L	6.2	K	none	Y
66	Ranjit Mondal	Hridaypur	WB	D	L	6.5	M	none	Y
67	Shyamal Das	Nabapally	WB	D	L	6.8	S	broker	Y
68	Tridib Roy	Ward 21	WB	D	L	7.1	K	none	M
69	Uday Ghosh	Barasat	WB	D	L	7.4	M	none	Y
70	Bhaben Halder	Jessore Road	WB	D	L	7.7	K	none	Y
71	Jaswinder Singh	Kharar	PB	P	S	1.6	S	own	M
72	Harpreet Kaur	Nabha	PB	P	S	1.7	B	none	Y
73	Balwinder Singh	Ghanaur	PB	P	S	1.8	B	none	Y
74	Gurmeet Singh	Rajpura	PB	P	S	2.0	B	broker	Y
75	Manpreet Kaur	Samana	PB	P	S	2.1	B	none	Y
76	Kuldeep Singh	Patiala	PB	P	S	2.2	B	own	Y
77	Parminder Kaur	Kharar	PB	P	S	2.3	B	broker	Y
78	Sukhwinder Singh	Nabha	PB	P	S	2.4	S	none	Y
79	Navjot Kaur	Ghanaur	PB	P	S	2.5	B	none	M
80	Amritpal Singh	Rajpura	PB	P	S	2.7	B	broker	Y
81	Gurpreet Singh	Samana	PB	P	S	2.8	B	own	Y
82	Simranjeet Kaur	Patiala	PB	P	S	2.9	B	none	Y
83	Harjinder Singh	Kharar	PB	P	M	3.2	S	own	M
84	Davinder Kaur	Nabha	PB	P	M	3.4	B	none	Y
85	Jagtar Singh	Ghanaur	PB	P	M	3.5	B	none	Y
86	Ranjit Singh	Rajpura	PB	P	M	3.7	B	broker	Y
87	Mandeep Kaur	Samana	PB	P	M	3.9	B	none	Y
88	Lakhwinder Singh	Patiala	PB	P	M	4.1	B	own	Y
89	Sandeep Kaur	Kharar	PB	P	M	4.3	B	broker	Y
90	Bhupinder Singh	Nabha	PB	P	M	4.5	S	none	Y
91	Gurmail Singh	Ghanaur	PB	P	M	4.6	B	none	M

#	Name	Place	St	How	Band	Ac	Fate	Phone	5d
92	Rajwinder Kaur	Rajpura	PB	P	M	4.8	B	broker	Y
93	Iqbal Singh	Samana	PB	P	L	5.3	S	own	M
94	Karamjit Kaur	Patiala	PB	P	L	5.7	B	none	Y
95	Avtar Singh	Kharar	PB	P	L	6.0	B	none	Y
96	Jasbir Kaur	Nabha	PB	P	L	6.3	B	broker	Y
97	Nirmal Singh	Ghanaur	PB	P	L	6.7	B	none	Y
98	Pawanpreet Kaur	Rajpura	PB	P	L	7.0	B	own	Y
99	Satnam Singh	Samana	PB	P	L	7.3	B	broker	Y
100	Tejinder Singh	Patiala	PB	P	L	7.7	S	none	Y

Table 5. Farmer book ($n = 100$).

9. Conclusions and impact of the activity

Across 395.0 acres (~790.0 t straw), last season failed in two modes: burn in Punjab and dump/mix in Madhyamgram and Barasat. Only 16 of 100 sold. Yet 81 of 100 would list if pickup were real - about 650.4 t and Rs 455,280, with ~975.6 tCO₂e estimated avoided if those lots utilise. That is the opening AgriNova is built to catch.

1. Home failure (Madhyamgram / Barasat) is dump and mix; Punjab failure is burn; shared cause is no plant number.
2. A clear majority (81/100) will list when five-day pickup is credible.
3. Pads take specified clean lots; they refuse mixed drain straw.
4. Ward desks will use utilised-vs-dumped aggregates; they will not put farmer phones on the screen.
5. Farmer-first carbon belongs on the same slip as moisture - on the road to India 2070.

Limitations: Punjab stratum is recall-based; civic notes are student conversations, not municipal circulars; Rs 700/t and 1.5 tCO₂e/t are working figures; carbon lines are simulators. Impact claimed is the interpreted n = 100 book plus a desk a judge can open at <https://agrinova-hazel.vercel.app>.

What we will not claim

We will not claim a state forecast, a certified carbon project, or an official municipality study. We will claim a representative 100-household book, a tested listing object, and a ward-scale pathway that is scientifically modest and locally usable.

10. Suggested system for abatement

AgriNova as a Madhyamgram / Barasat ward listing, with a phone listing for Punjab, wired to farmer-first estimated carbon and transparent utilised tonnes.

- Farm card: acres, crop, tonnes (2.0 t/acre default), moisture, neighbourhood; Bangla or Punjabi by place.
- Pad: moisture on slip; reject mixed drain waste; compost-first when wet; biomass-first when dry.
- Ward / SI: weekly utilised vs dumped - no KCC, Aadhaar, or farmer mobile.
- Logistics: trolley within five days; neighbourhood pool for small (S) lots.
- Carbon: estimated tCO₂e and credit simulator accrue first to the diverting farmer; labelled educational.

10.1 Improvements after field testing

- Wet / cannot-bale flag so Madhyamgram compost is offered before biomass.
- Small lots auto-join a lane pool.
- Conservancy weekly total of tonnes lifted - still without names.
- SMS listing for keypad phones; moisture written before the truck leaves.

10.2 Order of abatement

1. Farmer lists before the heap mixes or the window closes.
2. Score uses moisture so a wet lot is not sent to a dry-spec cabin.
3. Pickup date is visible on the same card.
4. Farmer-first estimated carbon is written when the lot is utilised.
5. Ward note records tonnes, not names.

11. Acknowledgements

We place on record our sincere gratitude to all who made this Form-A project possible.

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- Kendriya Vidyalaya Dum Dum, Kolkata - our school - for encouragement and the space to plan fieldwork.
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- Plant-floor contacts at the Madhyamgram compost pad and the Punjab biomass cabin.
- Dr. Bibartan Saha, Councillor, Ward 21, Barasat, and Shri Sudip Biswas, Sanitary Inspector, Madhyamgram.
- Our families, for Punjab phone introductions and travel support on local visits.
- Public sources - MoA&FW (NPMCR / CRM), MoEFCC (SWM Rules, 2016), CPCB / SAFAR, CACP, IARI / PAU - used as context.
- Any errors of interpretation remain our own as student investigators.

12. References

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- [5] Government of India statements on long-term net-zero / 2070 horizon (context).
- [6] SAFAR / IITM and CPCB episode notes on post-harvest PM2.5 (context only).
- [7] Commission for Agricultural Costs and Prices (CACP) - MSP cards for common paddy.
- [8] IARI / PAU residue-to-grain working bands (1.8-2.4 t/acre straw).
- [9] National Green Tribunal orders on crop-residue burning (enforcement context).
- [10] AgriNova live desk: <https://agrinova-hazel.vercel.app>
- [11] Kendriya Vidyalaya Dum Dum, Kolkata - institutional support for this Class XI project.

13. Declaration

We hereby declare that this Form-A project report on AgriNova is our own work as Class XI students of Kendriya Vidyalaya Dum Dum, carried out under the NCSC theme Science and Innovation for Sustainability. Survey sheets, analysis and the demonstration desk were prepared by us with guidance from our Guide Teacher. Carbon and price figures are working educational estimates, not certified carbon credits.

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Date _____